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1. Introduction:

This report focuses on analysing the 2011 UK Census dataset to extract meaningful insights using various analytical techniques. The task includes descriptive analytic, data visualization, classification on Approximate Social Grade column, regression on No of hours, applying associate rule mining and clustering on the given data. The results of each method will be presented, interpreted, and compared to fulfil the objectives of this analysis.

1. Basic Statistics

a. Importing the data

First import the required libraries and import the census data using pandas. Census data was provided on Moodle.

```
import numpy as np          # used for multidimensional arrays
import pandas as pd         # used for import the dataset
import matplotlib.pyplot as plt # used for visualising graphs
import seaborn as sns       # used for visualising graphs
#this will make your plot outputs appear and be stored within the notebook
%matplotlib inline
```

```
# Reading a CSV file named 'Census.csv' into a pandas DataFrame.
df = pd.read_csv('Census.csv')
```

b. Checking first and last 5 rows of data

Checking the first 5 rows and 5 last rows of data using `pd.head()` and `pd.tail()` method respectively as below:

```
# Display first 5 rows
df.head()
```

	Person ID	Region	Residence Type	Family Composition	Population Base	Sex	Age	Marital Status	Student	Country of Birth	Health	Ethnic Group	Religion	Economic Activity	Occupation	Industry
0	7394816	E12000001	H	2	1	2	6	2	2	1	2	1	2	5	8	2
1	7394832	E12000001	H	3	1	2	1	1	2	1	2	1	1	-9	-9	-9
2	7394719	E12000001	H	2	1	1	7	2	2	1	1	1	2	5	8	2
3	7394840	E12000001	H	1	1	2	6	4	2	1	3	1	2	5	9	5
4	7394711	E12000001	H	2	1	1	1	1	1	1	1	1	1	-9	-9	-9

```
# Display last 5 rows
df.tail()
```

click to scroll output; double click to hide

	Person ID	Region	Residence Type	Family Composition	Population Base	Sex	Age	Marital Status	Student	Country of Birth	Health	Ethnic Group	Religion	Economic Activity	Occupation	Ind
569735	7680019	E12000008	C	-9	1	1	2	1	2	2	1	1	2	1	3	
569736	7675184	E12000008	C	-9	1	1	3	1	2	2	2	2	1	1	1	
569737	7675176	E12000008	C	-9	1	2	5	2	2	1	1	1	2	2	1	
569738	7944614	W92000004	H	2	1	1	6	2	2	1	3	1	1	1	6	
569739	7945074	W92000004	H	1	1	1	5	4	2	1	2	1	1	1	5	

c. Checking shape and data types:

```
# Get shape of the DataFrame
df.shape
```

```
(569740, 19)
```

```
# Get data types of each column
df.dtypes
```

```
Person ID          int64
Region            object
Residence Type     object
Family Composition  int64
Population Base     int64
Sex               int64
Age              int64
Marital Status     int64
Student           int64
Country of Birth   int64
Health            int64
Ethnic Group       int64
Religion          int64
Economic Activity  int64
Occupation         int64
Industry           int64
Hours worked per week
No of hours        float64
Approximated Social Grade
dtype: object
```

Checking the data types and shapes as shown in the above figure:

It has 569740 rows and 19 columns of data and for the data types its two columns are objects, and the remaining are integers except column 'No of hours' which is float.

d. Statical Summary:

Data types were changed from numerical to categorical for required columns. The `df.describe(include = 'all')` function is used which provides statistical summary of a DataFrame as below:

Statistical summary for numerical column:

No of hours:

Stats	Value	Description
Count:	267419.00	The total number of observations in this column is 267,419.
mean	25.23	The average value of the data is 25.23.
Standard Deviation (std)	13.52	The measure of variability or dispersion in this column is 13.52.
Minimum (min)	1.00	The smallest value in this column is 1.00.
25th Percentile (25%)	27.00	Also known as the first quartile, 25% of the data falls below 27.00.
Median (50%)	37.00	The middle value or median is 37.00.
75th Percentile (75%)	45.00	Also known as the third quartile, 25% of the data falls below 45.00. Click or tap here to enter text.
Maximum (max)	60.00	The maximum value in this column is 60.00.

Statistical summary for categorical columns:

Region:

Count	Unique	Top	Freq	Description
569740	10	South East	88084	The dataset contains 569,740 non-missing data in the Region column, with 10 unique regions. The most common region is Southeast which appeared 88,084 times which makes it the most frequently occurring category.

Residence Type:

Count	Unique	Top	Freq	Description
569740	2	Not a communal establishment	559086	The Residence Type column has 569,740 non-missing data with 2 unique categories. The most common category is Not a communal establishment which appeared 559,086 times.

Family Composition:

Count	Unique	Top	Freq	Description
-------	--------	-----	------	-------------

569740	7	married	300961	The data shows that there are 569,740 entries with 7 unique family composition types. The most common family composition is marriage which appeared 300,961 times in that column.
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Population Base:

Count	Unique	Top	Freq	Description
569740	3	Usual resident	561039	There are 569,740 of not null data with 3 unique population bases. The most common category is Usual resident.

Sex:

Count	Unique	Top	Freq	Description
569740	2	female	289172	The data shows that there are 569,740 entries with 2 unique categories for sex. The most frequent category is female.

Age:

Count	Unique	Top	Freq	Description
569740	8	0 to 15	106832	There are 569,740 non missing values with 8 unique age groups. The most common age group is 0 to 15.

Marital Status:

Count	Unique	Top	Freq	Description
569740	5	single	270999	There are 569,740 non missing values with 5 unique marital status categories. The most frequent category is single with 270,999 entries.

Student:

Count	Unique	Top	Freq	Description
569740	2	No	443203	The data shows that there are 569,740 entries with 2 unique categories for student status. The most common category is No which appeared 443,203 times.

Country of Birth:

Count	Unique	Top	Freq	Description
569740	3	UK	485645	There are 569,740 non missing values with 3 unique categories for the column country of birth. The most common category is the UK which appeared 485645 times.

Health:

Count	Unique	Top	Freq	Description
569740	6	Very good health	264971	The data contains 569,740 non missing values with 6 unique health status categories. The most common category is

				Very good health which has appeared 264,971 times.
--	--	--	--	--

Ethnic Groups:

Count	Unique	Top	Freq	Description
569740	6	white	483477	The data includes 569,740 non missing values with 6 unique ethnic groups. The most common group is white, appearing 483,477 times.

Religion:

Count	Unique	Top	Freq	Description
569740	10	Christian	333481	The data consists of 569,740 non missing values having 10 unique religion categories. The most common category is Christian with 333,481 frequency.

Economic Activity:

Count	Unique	Top	Freq	Description
569740	10	Employee	216024	The data includes 569,740 non missing values with 10 unique categories of economic activity. The most common category is Employee appearing for 216,024 times.

Occupation:

Count	Unique	Top	Freq	Description
569740	10	People under age 16	149984	The data contains 569,740 non missing values with 10 unique occupation categories. The most frequent category is People under age 16 with 149,984 frequency.

Industry:

Count	Unique	Top	Freq	Description
569740	13	People under age 16	149984	The data contains 569,740 non missing values with 10 unique occupation categories. The most frequent category is People under age 16 with 149,984 frequency.

Hours worked per week:

Count	Unique	Top	Freq	Description
569740	5	People not working	302321	The data shows 569,740 non missing values with 5 unique categories for hours worked per week. The most common category is People not working which appears 302,321 times.

Approximate Social grade:

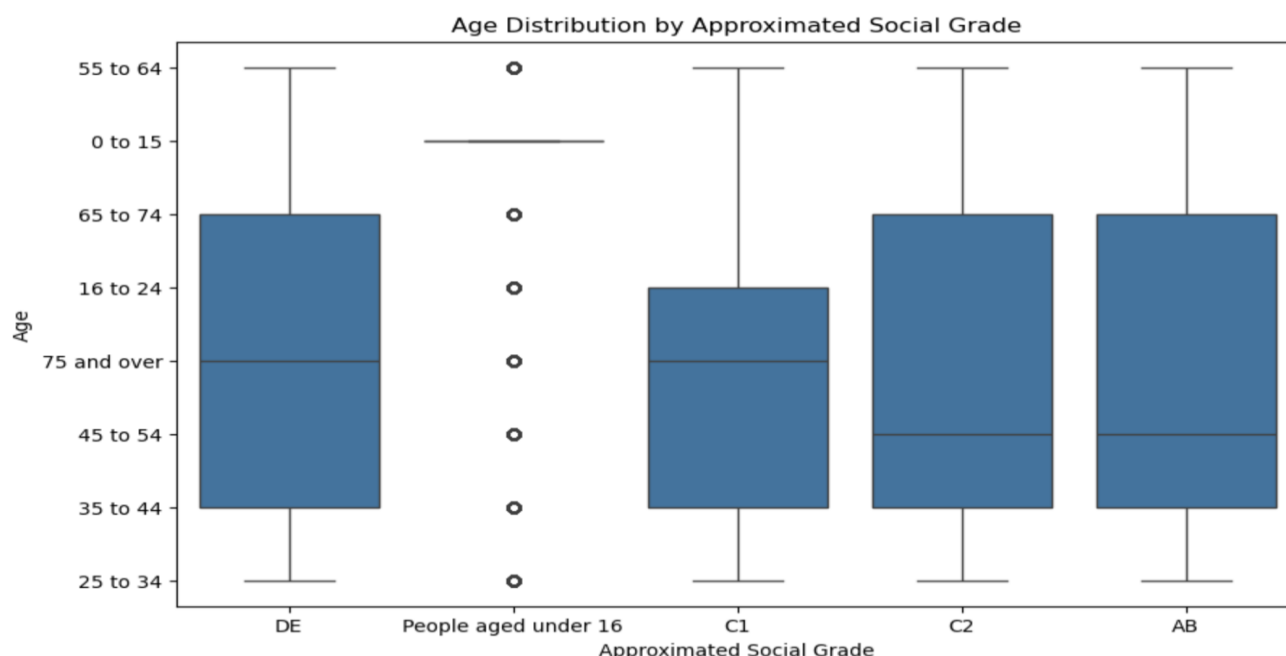
Count	Unique	Top	Freq	Description
569740	5	C1	159642	The data includes 569,740 non missing values with 5 unique social grade

				categories. The most common category is C1 with 159,642 frequency.
--	--	--	--	--

2. Data Visualization:

a. Box Plot:

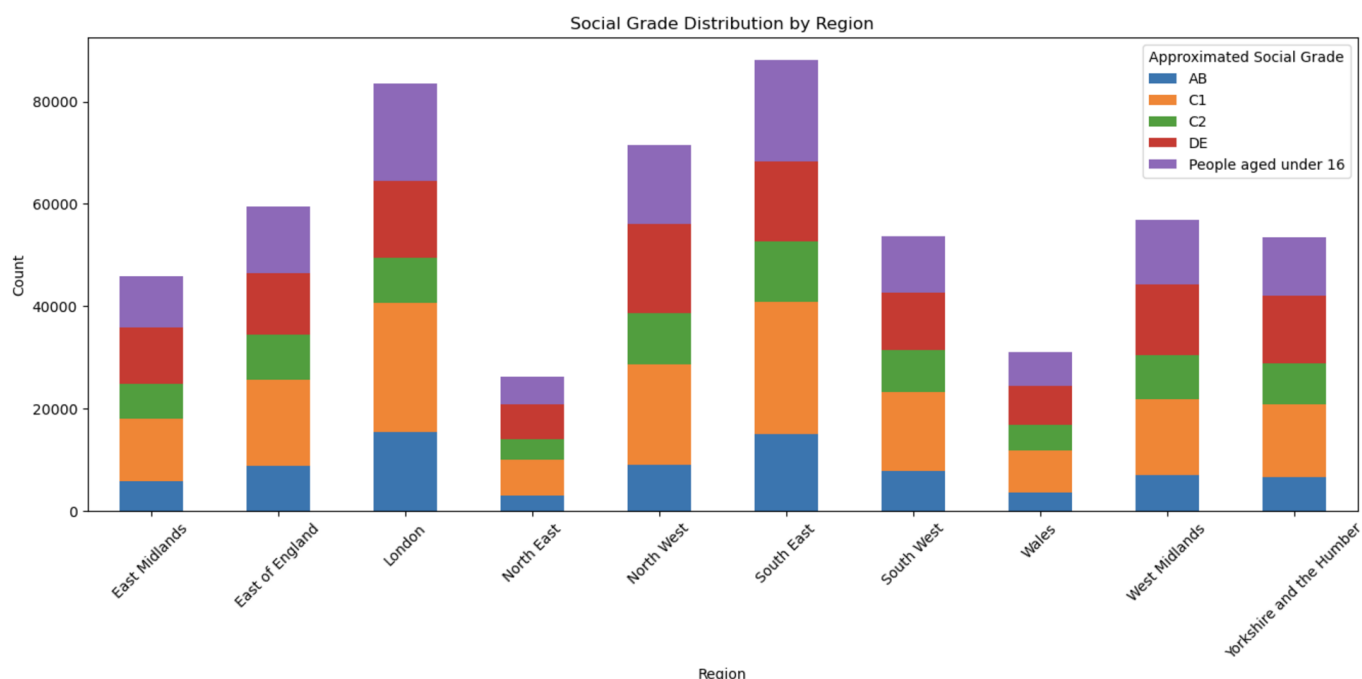
The age distribution by Approximate Social Grade is shown in the diagram below. There are eight age groups and five social classes. The social grade People aged under 16 has a narrow age range as expected, while grades such as AB, C1, and C2 show broader age distributions While the age group of Approximated Social Grade C1 is between 16 to 24 and 35 to 44, meaning that youth and adults persons belong under social group C1, the age group of Approximated Social Grade AB, C2, and DE are between 35 to 44, and 65 to 74 which means older people are belongs to this Approximate Social Grade.



b. Stacked Bar Plot:

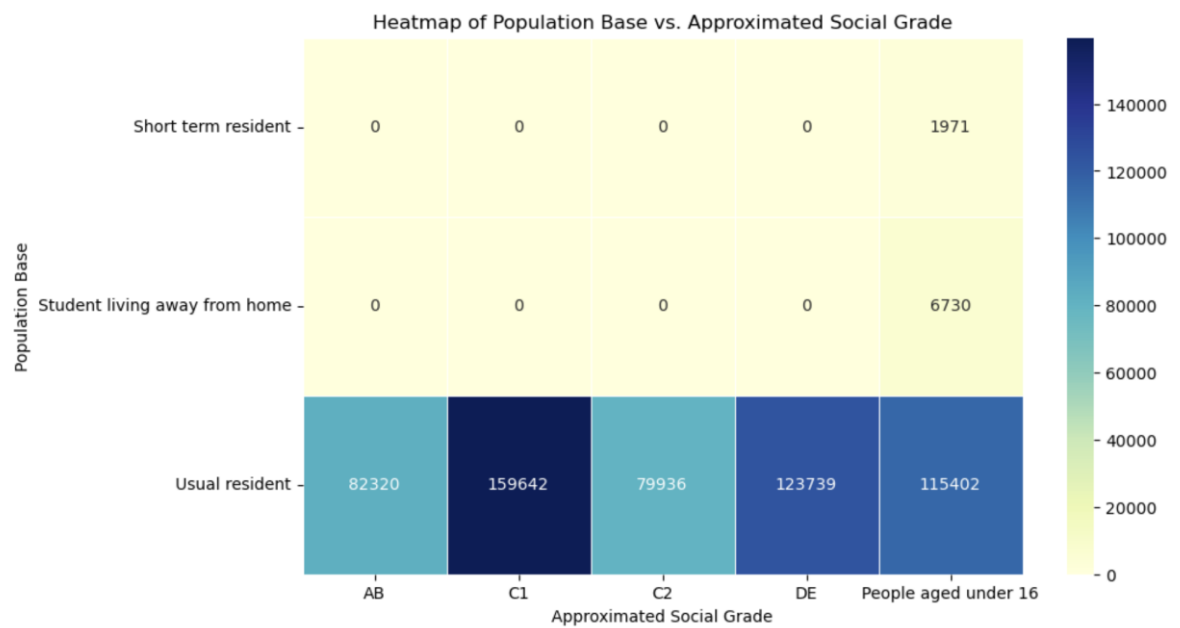
The chart below visualises the distribution of Approximated Social Grade across different Regions. It is stacked bar chart, where each region has a single bar split into different segments representing the categories of Approximated Social Grade. While the North East and Wales have a higher percentage of lower social grades (DE), London is more common in higher social grades (AB and C1) than other regions. The distribution of all classes in the South East is balanced. This graph did a good job by highlighting geographical differences in the composition of social grades. Each bar represents the total count of individuals for a region, broken down into segments (social grades). It seems that South East have the most people aged under 16 while least on North East. Most of the

people are under social group of C1 and Least people are living in the North East. Overall least people are in approximate social group C2.



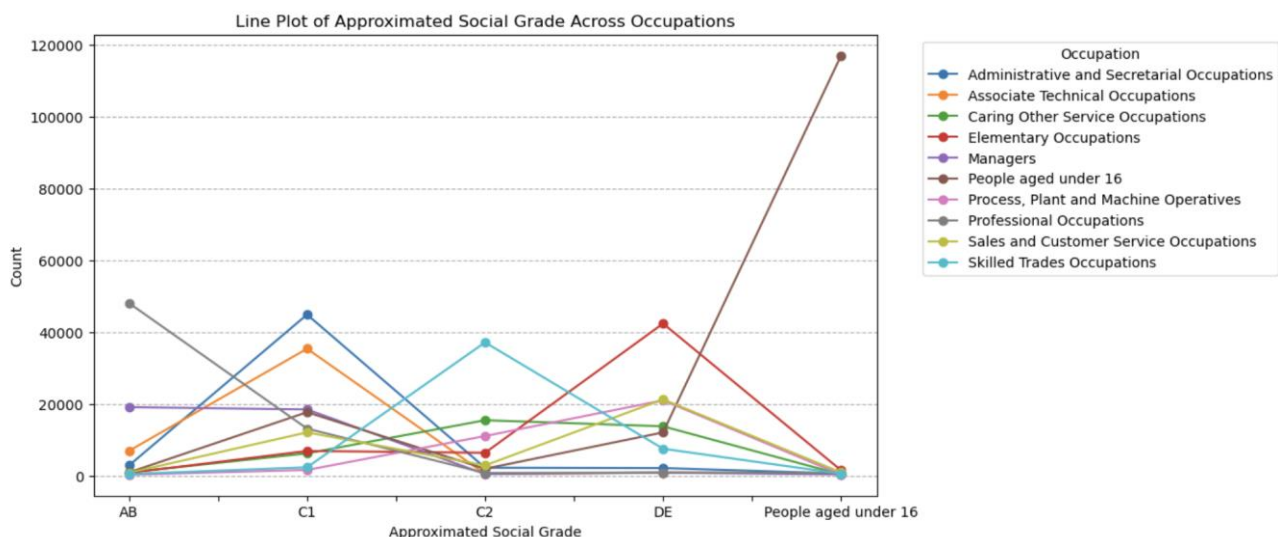
c. Heatmap:

The diagram below shows the heatmap of Population Base vs Approximate Social grade. This heatmap shows how the population base is distributed across approximate approximated social grades, such as 'AB,' 'C1', 'C2', 'DE', and 'People aged under 16.' Three categories are used to categorize the population: 'Short-term resident', 'Student living away from home,' and 'Usual resident.' With a peak of 159,642 in the 'C1' grade, the 'Usual resident' category has the greatest counts of any social grade. With a maximum of 6,730 in the 'People aged under 16' group, the numbers for the 'Student living away from home' category are significantly less. In all social grades, the 'Short-term resident' category has zero values. The heatmap's colour gradient, where darker colours denote higher counts, graphically highlights the majority of regular inhabitants.

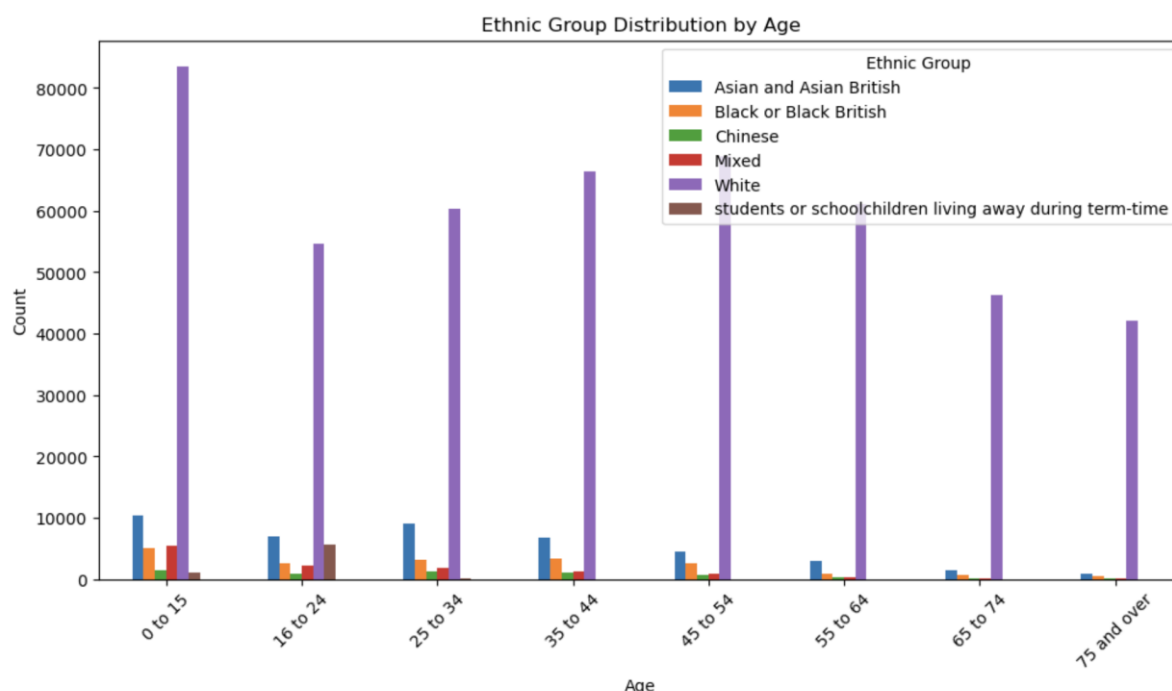


d. Line Plot:

The chart below shows the line plot of Approximate Social Grade Across Occupations. There are 10 different occupation which are plotted across 5 different social grades. In the line graph, different occupations are distributed throughout approximated social grades, such as 'AB,' 'C1,' 'C2,' 'DE' and 'People aged under 16.' The social grades are shown on the x-axis, while the number of people is shown on the y-axis. As shown in the legend, each occupation is represented by a different coloured line, such as 'Managers,' 'Skilled Trades Occupations,' 'Sales and Customer Service Occupations,' and so on. There are clear differences in the figure, with certain professions peaking at popular economic statuses. For instance, for at least one occupation, the social grade 'DE' has the highest count, whereas other grades, such as 'AB' and 'C1,' displayed lower distributions. The association between social grades and occupation types is clearly shown in the chart.



e. Clustered Bar Plot:



The figure above is the cluster bar chart of Ethnic Groups Distribution by Age. The distribution of various ethnic groups across age groups is shown in the above chart. Age groupings, from '0 to 15' to '75 and over' are represented by the x-axis, while the number of people is displayed on the y-axis. The legend shows the color-coded ethnic categories, which include White, Asian and Asian British, Black or Black British, Chinese, Mixed, and Students or schoolchildren living away during term time. All age categories are dominated by the White group, which has the highest numbers, especially in younger age ranges. The numbers for other ethnic groupings, like Asian and Asian British and Black or Black British are more stable and smaller. Students or schoolchildren living away from home are quite common in the age group '16 to 24' which is evidence of their migration for educational reasons. The distribution tends to decrease as people get older, with fewer persons falling into the '75 and over' category.

3. Data Preprocessing

df.info()				df.isnull().sum()	
<class 'pandas.core.frame.DataFrame'>				Person ID	0
RangeIndex: 569740 entries, 0 to 569739				Region	0
Data columns (total 19 columns):				Residence Type	0
#	Column	Non-Null Count	Dtype	Family Composition	0
0	Person ID	569740 non-null	int64	Population Base	0
1	Region	569740 non-null	object	Sex	0
2	Residence Type	569740 non-null	object	Age	0
3	Family Composition	569740 non-null	int64	Marital Status	0
4	Population Base	569740 non-null	int64	Student	0
5	Sex	569740 non-null	int64	Country of Birth	0
6	Age	569740 non-null	int64	Health	0
7	Marital Status	569740 non-null	int64	Ethnic Group	0
8	Student	569740 non-null	int64	Religion	0
9	Country of Birth	569740 non-null	int64	Economic Activity	0
10	Health	569740 non-null	int64	Occupation	0
11	Ethnic Group	569740 non-null	int64	Industry	0
12	Religion	569740 non-null	int64	Hours worked per week	0
13	Economic Activity	569740 non-null	int64	No of hours	302321
14	Occupation	569740 non-null	int64	Approximated Social Grade	0
15	Industry	569740 non-null	int64		
16	Hours worked per week	569740 non-null	int64		
17	No of hours	267419 non-null	float64		
18	Approximated Social Grade	569740 non-null	int64		
dtypes: float64(1), int64(16), object(2)					
memory usage: 82.6+ MB					

- Data Cleaning:** Most of the data consists of integers, however the columns Region and Residence Type contain object data types. The graphic on the upper right illustrates the total of the null values. The dataset contains a total of 569,740 rows, of which 302,321 contains null values in the 'No of hours' column, representing approximately 53% of the data. So, if we remove half of the data, it may lose its characteristics; thus, we select the median to fill the null values. Unlike the mean, the median is not skewed by unusually high or low values, making it suitable for skewed distributions. Census data consists of outlier -9 which is replaced by 0. The columns 'Person ID' is not required so we remove it.
- Label Encoding:** The fit_transform method is applied to the Residence Type and Region columns of a DataFrame. This method learns the unique categories in each column and assigns them integer labels (0 and 1).
- Scaling the numeric data:** Scaling the column 'No of hours' which is helpful for algorithms that are sensitive to the data.
- Sampling the data:** As the data is too large it is computational expensive so taking the sample of the data. Randomly selecting 10% of the rows from the DataFrame df to create a new sample, df_sample .
- Splitting the data:** Separating the predictor attributes and the target attributes then splitting the data. Splitting the 75 % of data for training and remaining 25 % of the data for testing purpose.

```
X = df_sample.iloc[:, :-1].values # predictor attributes
y = df_sample.iloc[:, -1].values # target attribute

from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_state=0)
```

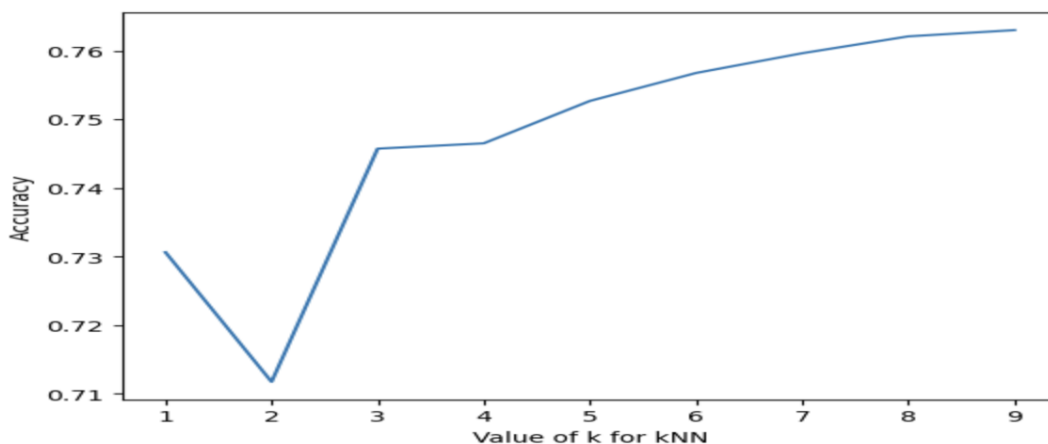
4. Classification:

For classification KNN, Naive Base, MLP classifier, Boosting and XGBoost were used. Summary of all the models are as follows:

a. KNN:

The KNN model performs well, with 75% accuracy, an AUC of 0.92, and low regression errors (MAE: 0.33, MSE: 0.54, RMSE: 0.74). It performs well on classification tasks due to its great class separability and reliable precision, recall, and F1-scores.

Plotting the line graph as shown below after calculating the accuracy score using various k values between 1 and 10. With a k value of 8, the graph indicates great accuracy.



The KNN model obtains 76% accuracy and low regression errors (MAE: 0.31, MSE: 0.51, RMSE: 0.71) after cross-validation of 10 folds passing full data, which is a slight improvement over the previous results. It showed great class separability and efficient performance with 12,071 accurate predictions for the first class and consistent precision, recall, and F1-scores.

b. Naive Bases:

The Naive Bayes classifier achieves 44% accuracy, an AUC of 0.84, and regression errors of MAE: 0.84, MSE: 1.55, and RMSE: 1.24 which is comparatively poorer result than KNN. With 513 correct predictions for the first class, it demonstrated strong class separability but poor overall classification performance.

After cross-validation of 10 folds, passing full data, its performance was similar which is poorer than KNN. the Naive Base model achieves 44% accuracy, and high regression errors (MAE: 0.84, MSE: 1.54, RMSE: 1.24) which are similar to before. With 2,138 correct predictions for the first class and uniform precision, recall, and F1-scores, it showed weak class separability and effective performance.

c. MLP Classifier:

The MLP Classifier achieves 80% accuracy, an AUC of 0.96, and regression errors of MAE: 0.25, MSE: 0.38, and RMSE: 0.62 which is an impressive performance. With 3195 correct predictions for the first class, it showed solid class separability and low prediction error, its classification accuracy is far better than KNN and Naive Base.

After cross-validation of 10 folds passing full data, the MLP model achieves 81% accuracy, and low regression errors (MAE: 0.22, MSE: 0.35, RMSE: 0.59) which are slightly improved from before. With 3195 correct predictions for the first class and consistent precision, recall, and F1-scores, it showed strong class separability and effective performance.

d. Bagging:

With 80% accuracy and an AUC of 0.94, the Bagging model performs satisfactorily and has good class separability. With strong precision, recall, and F1-scores across classes, it showed good classification performance and minimal regression error (MAE: 0.26, MSE: 0.40, RMSE: 0.63). When everything is considered, the model performs well in categorization.

After cross-validation passing full data, the Bagging model achieves 80% accuracy, and low regression errors (MAE: 0.25, MSE: 0.41, RMSE: 0.64) which are slightly improved from before. With 12,619 correct predictions for the first class and consistent precision, recall, and F1-scores, it demonstrates strong class separability and effective performance.

e. XGBoost :

The XGBoost model has good class separability with an 83% accuracy rate and a high AUC score of 0.97, demonstrating impressive performance across a range of measures. The classification performance is strong, with good precision, recall, and F1-scores, and the regression error is minimal (MAE: 0.22, MSE: 0.34, RMSE: 0.59). These findings imply that the XGBoost model has excellent classification performance and is highly effective.

After cross-validation passing full data, the XGBoost model achieves 83% accuracy, and low regression errors (MAE: 0.21, MSE: 0.34, RMSE: 0.58) which are slightly improved from before. It showed great class separability and efficient performance with 12,638 accurate predictions for the first class and consistent precision, recall, and F1-scores.

f. Comparision:

models	Ac cur acy	AUC	MAE	MSE	RMSE	Key observation

	in %					
KNN	75	0.92	0.33	0.54	0.73	- Accuracy: 75% - AUC: 0.92 (have good separability ability) - Errors: Have moderate MAE (0.33), MSE (0.54), and RMSE (0.73).
Naive Bases	44	0.84	0.84	1.55	1.24	- Accuracy: 44% (have lowest accuracy among the models) - AUC: 0.84 (poorer ability to distinguish between classes) - Errors: Highest MAE (0.84), MSE (1.55), and RMSE (1.24), which indicates the poor predictions.
MLP Classifier	80	0.96	0.25	0.38	0.62	- Accuracy: 82% - AUC: 0.96 (have strong separability ability) - Errors: Having MAE (0.25), and low MSE (0.38) and RMSE (0.62), indicating excellent performance.
Bagging	80	0.94	0.26	0.40	0.63	- Accuracy: 80% - AUC: 0.94 (good separability ability) - Errors: Slightly higher MAE (0.26), MSE (0.40), and RMSE (0.63) compared to MLP Classifier and XGBoost.
XGBoost	83	0.97	0.22	0.34	0.59	- Accuracy: 83% (highest among all models) - AUC: 0.97 (best ability to distinguish between classes) - Errors: Lowest MAE (0.22) with MLP Classifier, with the best MSE (0.34) and RMSE (0.59) values.

Conclusion Of Classification models:

- **Best Overall Model:** Because of highest accuracy, best AUC, and minimal errors, **XGBoost** is best classification for the given data.
- **Worst Model:** **Because** of lowest accuracy, low AUC, and high errors, **Naive Base** is worst classification for the given data.

5. Regression:

For regression I have used 3 models that are Linear Regression, CatBoost and Decision Tree Regression, 75% of the data were taken for the training while remaining 25% of the data were taken for testing purpose. The performance metrics of models are as follows:

a. Linear Regression:

The linear regression model has impressive performance. The Mean Absolute Error (MAE) of 4.07 indicates a minimal average prediction error. The Mean Squared Error (MSE) is 22.57, and the Root Mean Squared Error (RMSE) is 4.75, indicating that while bigger errors are infrequent, they do exist. The r^2 score of 0.876 signifies that the model accounts for about 87.6% of the variance in the target data, demonstrating robust predictive accuracy.

b. CatBoost Regressor:

The performance metrics demonstrate that the CatBoost regressor showed outstanding results, with a Mean Absolute Error (MAE) of 4.08, indicating a negligible average prediction error. The Mean Squared Error (MSE) of 22.57 and the Root Mean Squared Error (RMSE) of 4.75 indicate large differences linked to this method. The model explains roughly 87.6% of the variance in the target variable, showing a significant level of predictive accuracy.

The model was refined utilizing the "RMSE" loss function, and logging was deactivated for simplicity. A range of values for iterations, learning rate, and depth was done using a parameter grid. In order to find the optimal configuration, we used GridSearchCV, which uses parallel processing to increase efficiency, to evaluate these combinations on the training data (X_{train} and y_{train}). However, even throughout the tuning process, the model's performance showed negligible improvement and remained same.

c. Decision Tree regression:

The performance metrics indicate that the Decision Tree regressor model performs well. The Mean Absolute Error (MAE) of 4.08 shows that the average prediction error is low. The Mean Squared Error (MSE) of 22.57 and the Root Mean Squared Error (RMSE) of 4.75 shows that few errors are present. The r^2 score of 0.876 indicates that the model explains approximately 87.6% of the variance in the target variable, showing strong predictive accuracy.

Comparison:

Models	MAE	MSE	RMSE	R2_score	Key performance
Linear Regression	4.07	22.57	4.75	0.87	Lowest MAE (4.07) among the models, showing slightly better error minimization. Performs equally well on variance explanation with an r2_score of 0.87.
catBoost	4.08	22.57	4.75	0.87	Matches Linear Regression in terms of MSE, RMSE, and r2_score (0.87). Slightly higher MAE (4.08), but the difference is negligible.
Decision Tree	4.08	22.57	4.75	0.87	Identical performance to CatBoost in all metrics. Good option if interpretability and feature importance are priorities.

Conclusion:

All three models performed similarly in terms of error and variance explained. If computation time or model interpretability is important then Linear Regression is an excellent choice due to its simplicity.

6. Rule Mining:

For rule mining I have used apyori and Fpgrowth algorithms.

a. Apyori rule mining;

Rules	Antecedent	Consequent	Support	Confidence	Lift	Description
1	People not working, not a communal establishment	Christian, UK	0.25	0.60	2.13	People who are not working and not living in communal establishments are more likely to be Christian and living in the UK, with a confidence of 0.60 and lift of 2.13.
2	Christian, UK, usual resident	not a communal establishment, white	0.25	0.90	2.12	Christians who live in the UK and are usual residents are highly likely to not live in communal establishments and to be white

						with confidence 90% and lift of 2.12.
3	UK, no, people not working	Usual resident, white	0.27	0.95	2.12	People in the UK who are not student and working are likely to be usual residents and white with confidence 95% with lift 2.12.
4	UK, no, not a communal establishment , usual resident	White,37.0	0.26	0.95	2.12	People in the UK who are not in communal establishments, are usual residents, and are working 37.0 hour per week are associated with being white with confidence 95% with lift 2.12.
5	married	white	0.46	0.84	0.99	Married people are likely to be white, with confidence 84% and lift of 0.99.

b. Fpgrowth rule mining:

Rule	Antecedent	Consequent	Support	Confidence	Lift	Description
1	Not a communal establishment	Usual resident	0.96	0.98	1.0	Most people who are not in communal establishments are likely to be categorized as Usual residents. The confidence of 98% and support of 96% suggest a strong relationship, with lift being 1.0.
2	White	Usual resident	0.84	0.99	1.01	Individuals who are identified as White have a 99% probability of being Usual residents having the support of 84% and confidence of 99% indicate a strong association, with a lift of 1.01.
3	Uk, usual resident	Not a communal establishment	0.84	0.98	1.08	Individuals who are both from the UK and classified as Usual residents are highly likely to not be in

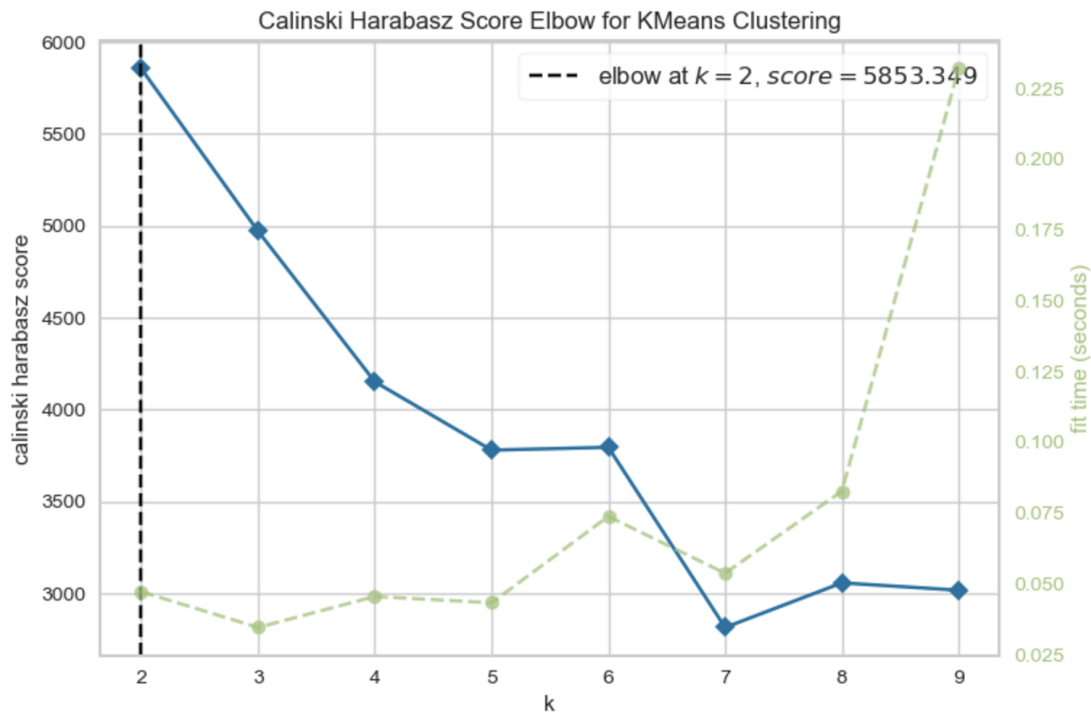
						communal establishments with confidence of 98% and life of 1.08.
4	Uk	White	0.79	0.92	1.08	Person from the UK has a 92% chance of being classified as "White." The support (79%) and confidence (92%) reflect a strong correlation, while the lift value of 1.08 indicates that this relationship is stronger.
5	Not a communal establishment, Usual resident	UK, White	0.77	0.80	1.01	Individuals who are not in communal establishments and are Usual residents are likely to be from the UK and classified as White. With a support of 77% and a confidence of 80% with lift of 1.01.

7. Clustering:

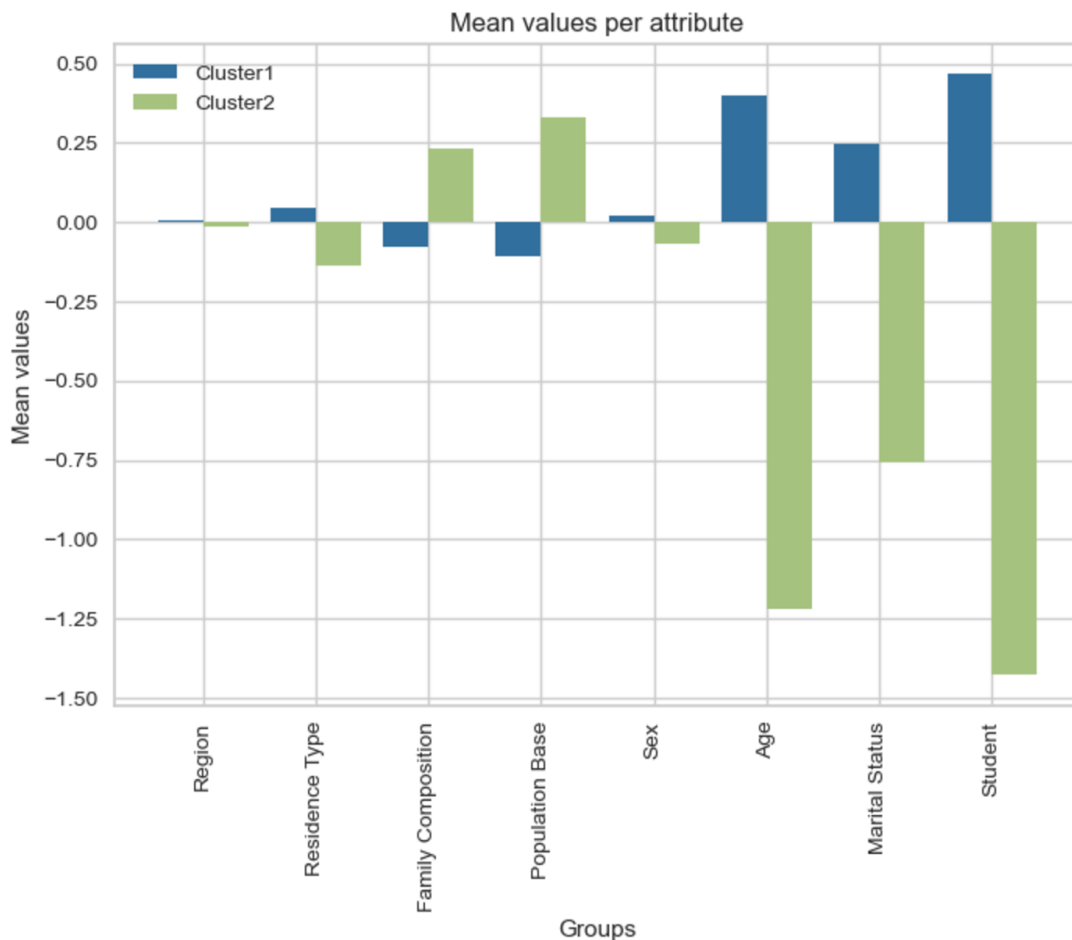
For clustering I have used k – means, Hierarchical Clustering: and Gaussian Mixture Model (GMM);

a. K – means:

For k means algorithms I assumed number of clusters as 3 at first then finding the centre of the clusters, but for finding the optimal number of clusters I used elbow method and visualize it with yellow brick using calinski_harabasz metric as follow:



Basically, this plot suggest that the optimal number of clusters must be 2, then data is scaled and fitted the data into cluster 1 and cluster 2 for finding the mean of both cluster which is then added in the data frame. Then data is divided into two parts for visualization. For this plot first 8 columns were used.



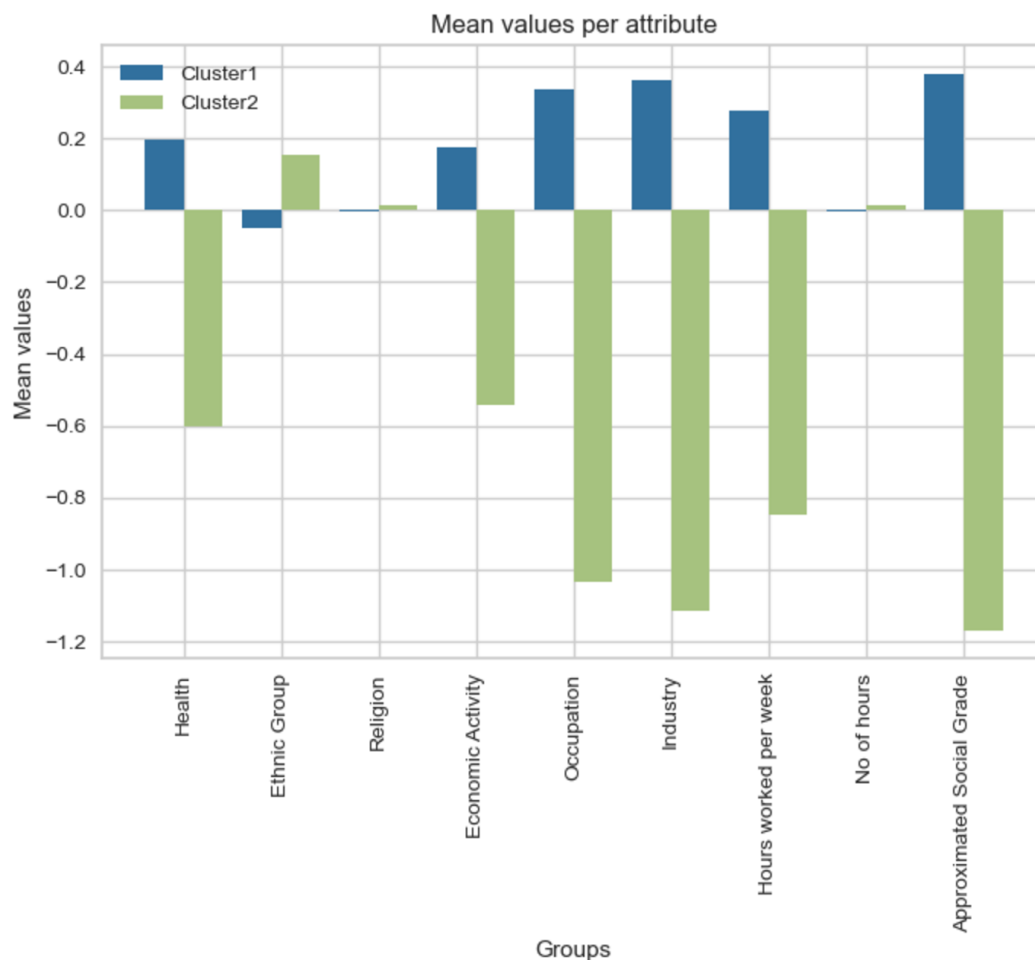
Observations:

1. Cluster 1 (blue bars):

- Shows positive mean values for attributes like Age and Marital Status. Negative mean values for Sex, Resident Type, and Student.

2. Cluster 2 (green bars):

- Shows opposite trends to Cluster 1, with negative mean values for Age and Marital Status and positive mean values for Sex and Student.



Observations:

1. Cluster 1 (blue bars):

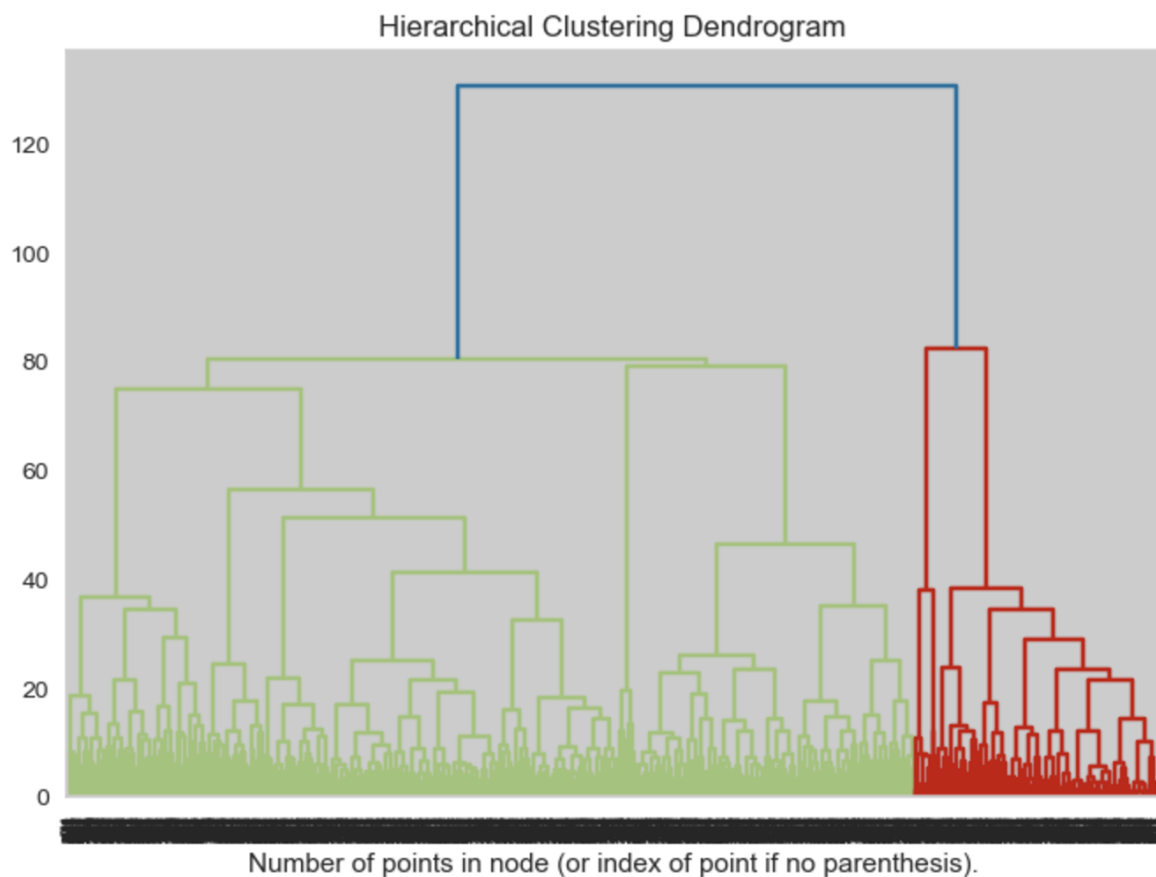
- Positive mean values are observed for columns like Economic Activity, Occupation, Industry and Hours Worked per week.
- Negative values for columns such as Ethic Group and Religion.

2. Cluster 2 (green bars):

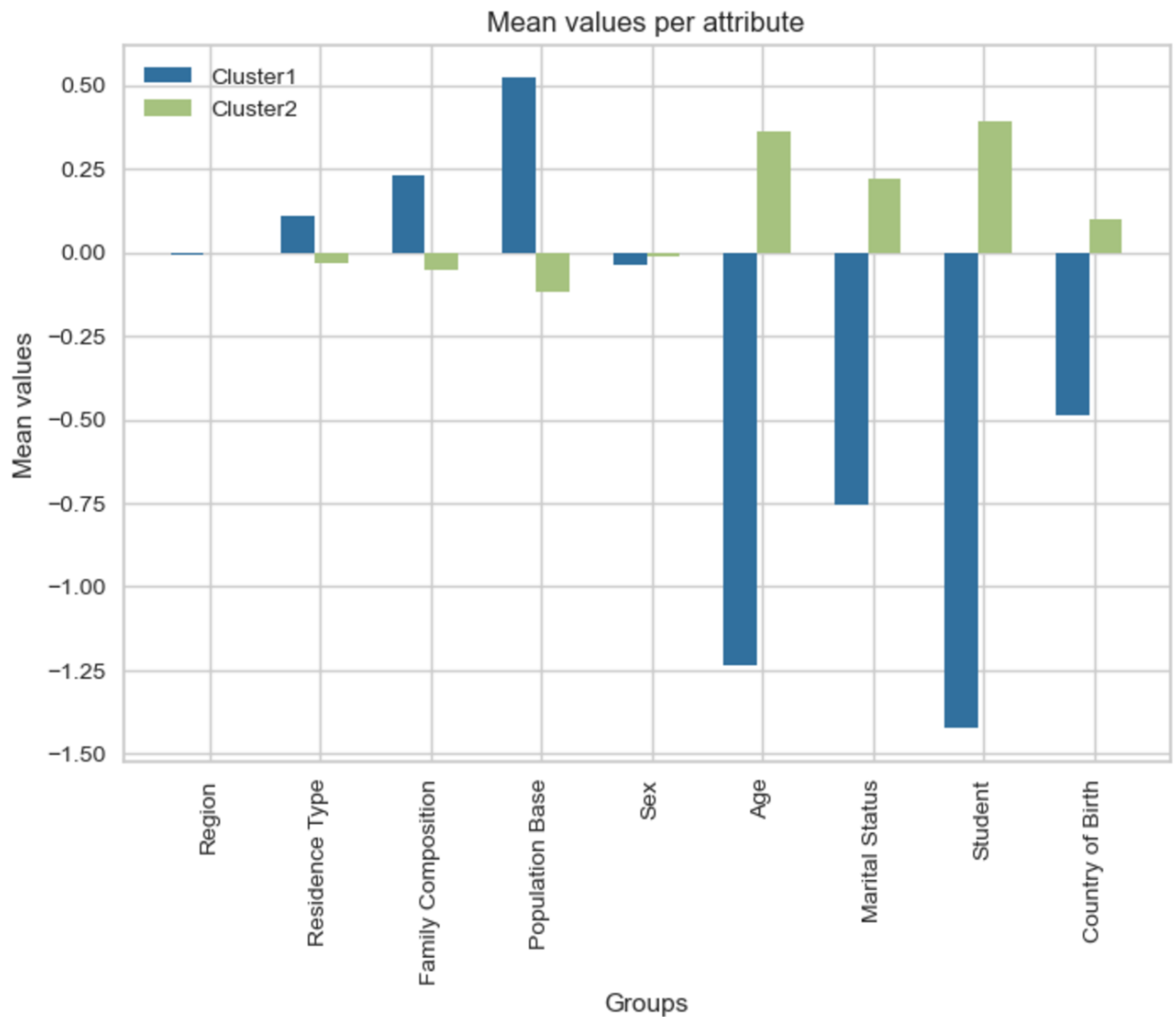
- a. Strongly negative mean values are observed for Hours Worked per week, Numbers of hours and Approximated Social grade.
- b. Positive mean values are seen for column Health Ethnic Group and Religion.

b. Hierarchical Clustering:

It is second algorithms used for clustering. For this model data is randomly selected from `scaled_features_df` without replacement. Then dendrogram is plotted as below:



Which suggest being two optimal number of clusters then two clusters that are cluster 1 and cluster are plotted as below:



Observations:

Cluster 1 (blue bars):

- The age and student means were strongly negative. Based on age, it appears that Cluster 1 consists mainly of younger persons. Additionally, the fact that it is less likely to contain students further supports this assumption.
- Positive mean values for region and marital status which mean individuals in Cluster 1 are more likely from certain regions and have specific marital status characteristics.

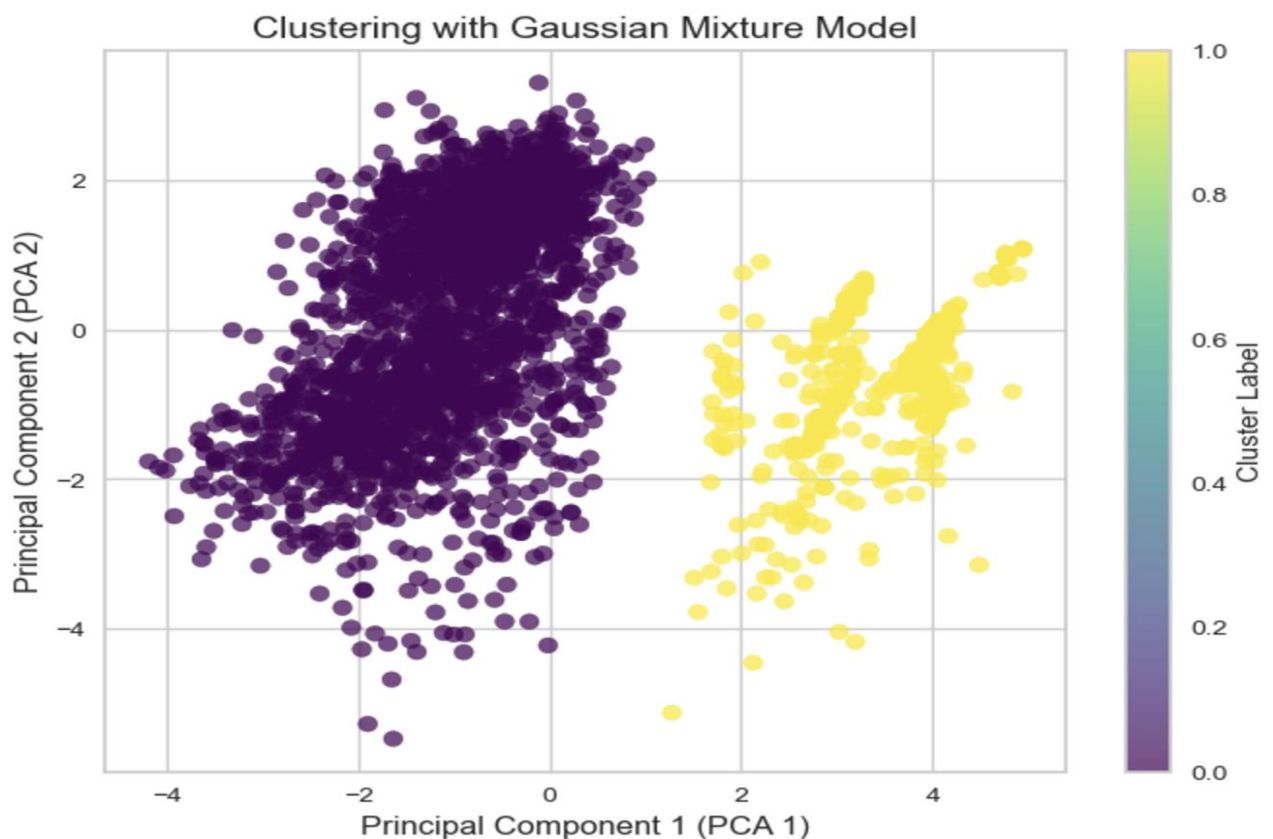
Cluster 2 (green bars):

- The results for both age and student show very positive means. Students are more likely to be located in this cluster if their age is an indicator of their dominance.
- Slight positive values for Country of Birth which may relate to people being born in specific regions or countries.

c. Gaussian Mixture Model (GMM):

Unlike k-means, the Gaussian combination Model (GMM) can handle clusters of different forms and sizes since it uses the assumption that the data is created from a combination of Gaussian distributions. The data was first scaled using StandardScaler. Then, using GMM, it was grouped into two groups, with `n_components=2`. Each cluster is represented by a Gaussian distribution, complete with its own mean and covariance matrix. The dataset was reduced to two parts for visualization using. The clusters were then shown in a two-dimensional scatter plot, with colours assigned according to the expected cluster names. For analysing

overlapping groupings, GMM's probabilistic nature enables soft clustering, where each data point is given to the cluster with the highest likelihood.



Conclusion: For clustering three algorithms were used that are K – means clustering, Hierarchical clustering, and Gaussian Mixture Model (GMM). All their models predicted clustered to be 2. For k – means elbow methods is used to find optimal number of clusters

using calinski_harabasz metric and dendrogram for Hierarchical clustering. As all three clusters were good at prediction but K – means is better for clustering because of its simplicity and can handle large numbers of datasets.

8. Reference:

- Scikit-learn developers. (n.d.). *Scikit-learn: Machine learning in Python*. Retrieved November 20, 2024, from <https://scikit-learn.org/stable/>
- University of Portsmouth. (n.d.). *Regression and association rules*. Retrieved November 20, 2024, from https://moodle.port.ac.uk/pluginfile.php/4959547/mod_resource/content/3/RegressionAndAssociationRules.pdf